Miguel Zumalacárregui

Ramón y Cajal Fellow — Instituto de Física Teórica IFT-UAM/CSIC, Madrid
Group Leader, ACR Department — Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam

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The publication record is maintained on INSPIRE-HEP; research projects are described on the website.

Positions

- **Ramón y Cajal Fellow**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2026–
- **Group Leader**, Astrophysical and Cosmological Relativity
  Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam · 2020–
- **Marie Skłodowska-Curie Global Fellow**
  Berkeley Center for Cosmological Physics, with return phase at IPhT Saclay · 2017–2020
- **Nordita Fellow**
  Nordic Institute for Theoretical Physics, Stockholm · 2015–2017
- **Postdoctoral Researcher**
  Institut für Theoretische Physik, University of Heidelberg · 2013–2015
- **Postdoctoral Researcher**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2013

Education

- **PhD in Physics**, *Cum Laude* with International Mention
  Universidad Autónoma de Madrid · 2012 · Graduate work at ICC Barcelona and IFT-UAM/CSIC (FPI fellowship)
  Dissertation: Probing the Foundations of the Standard Cosmological Model · defense presentation
- **Master in Theoretical Physics**
  Universidad Autónoma de Madrid · 2009
  Thesis: Disformal Quintessence: Observational Constraints and Linear Perturbations

Supervision

Postdoctoral researchers

- **Daniel Ballard** — from November 2026
- **Han-Gil Choi** — 2026–
- **Xikai Shan** — Affiliated postdoctoral researcher, 2026–
- **Juan Urrutia** — from November 2026
- **Héctor Villarrubia-Rojo** — 2026–
- **Srashti Goyal** — 2023–2026 → Independent Fellow at IUCAA
- **Serena Giardino** — 2024–2025 → Postdoc at Imperial College London
- **Guilherme Brando** — Humboldt Fellow, 2022–2024 → Tenured Researcher at UFES, Brazil
- **Giovanni Tambalo** — 2020–2023 → Postdoc at ETH Zürich, then Marie Skłodowska-Curie Fellow at IPhT Saclay

PhD students

- **Emanuela Alberici** — from October 2026

- **Manon van Zyl** — from October 2026
- **Nicola Menadeo** — PhD student, U. Naples (co-supervised), 2023–2025, long-term stay → Postdoc at Humboldt University of Berlin
- **Stefano Savastano** — PhD student (co-supervised with A. Buonanno), 2020–2024 → PhD awarded at HU Berlin
- **Carlos García-García** — PhD student, IFF-CSIC (co-supervised), 2017–2020 → Beecroft Fellow at Oxford, now tenure-track scientist at CIEMAT

Master's students

Matteo Pagani (U. Bologna, co-supervised, 2026); **Louis Salimbeni** (ETH Zürich, co-supervised, 2026); **Pablo Vega del Castillo** (LMU, 2026); **Shashwat Singh** (Paris Observatory, 2022–2023 → PhD at the University of Glasgow); **Julius Streibert** (FU Berlin, co-supervised, 2022–2023); **Richard Stiskalek** (LMU, co-supervised, 2021–2022 → PhD at Oxford, pre-doctoral Fellow at CCA, now postdoc at Oxford); **Alessandro Longo** (U. Trieste, co-supervised, 2020–2021 → PhD student at SISSA); **Janina Renk** (U. Heidelberg, co-supervised, 2015–2016 → PhD at Stockholm University).

Research interns and undergraduates

Selin Üstündag (2026 → PhD at Newcastle University); **Marcel Glätzer** (HU Berlin, 2024–2025); **Jaime Raynaud** (TU Berlin & KTH, 2024–2025 → PhD at MPIA); **Vaibhav Omble** (LMU, 2023–2024); **Daryna Yushchenko** (Princeton summer program, 2024 → PhD at Carnegie Mellon University); **Lyla Choi** (Princeton summer program, 2023 → PhD at Stanford); **Yukei Murakami** (UC Berkeley, 2018–2019 → PhD at Johns Hopkins University); **Brindha Kanniah** (MIT summer program in Heidelberg, 2015 → Master's at MIT).

Visiting researchers, and students mentored informally rather than formally supervised, are listed at miguelzu-malacarregui.es/group.html.

Scientific coordination

- **LISA Consortium**
  member since 2018
  Council member and Senior Co-Chair of the Cosmology Working Group (2025–); section coordinator for gravitational-wave lensing in the LISA Cosmology white paper (2022)
- **Euclid Consortium**
  member since 2014
  Co-lead of the Science Working Group on Gravitational Waves and of its "bright sirens" work package (2025–); member of the Strong Lensing, Cosmology Theory, and Supernovae & Transients working groups
- **GW-Space 2050**
  Coordinator for the cosmography section, defining ESA's gravitational-wave missions after LISA (2023–)
- **Multi-messenger lensing white paper**
  Coordinator for the microlensing section (2024)

Workshops and meetings organized

- 13th LISA Cosmology Working Group Workshop
  Barcelona, 2026
- 12th LISA Cosmology Working Group Workshop
  Tallinn, 2025
- Lensing and Wave Optics in Strong Gravity
  Erwin Schrödinger Institute, Vienna, 2024

- LISA joint Fundamental Physics + Waveform meeting
  Potsdam, 2024
- ACR Division Seminars
  Max Planck Institute for Gravitational Physics, 2020–2022
- BCCP Cosmology Lunch
  UC Berkeley, 2016–2017
- Cosmology Seminars, Institut für Theoretische Physik
  University of Heidelberg, 2014–2015
- TRR33 Winter Schools
  Passo del Tonale, 2013, 2014 and 2015

Teaching

Graduate schools

- **Gravitational waves**, TAE (Taller de Altas Energías), Benasque, 2024
  Slides
- **Tests of gravity and dark energy with cosmology and gravitational waves**
  II Mesoamerican Workshop on Gravity and Cosmology, Chiapas, 2018
- **Alternative gravity theories**, Cosmology School in the Canary Islands, 2017
  Video lecture
- **School for Cosmology Tools**, IFT Madrid, 2017
  Lectures on CLASS, hi_class and MontePython; material on the Talks & Teaching page
- **Workshop on CLASS and MontePython**, MPIA Munich, 2014

University courses

- **Berkeley Connect Instructor**, UC Berkeley, 2018–2019
  Two semesters of undergraduate mentoring and teaching
- **Lecturer**, University of Heidelberg, 2014–2015
  Advanced Cosmology; Introduction to CMB Physics; Statistical Mechanics; advanced seminar
- **Teaching assistant**, University of Heidelberg, 2014–2015
  Theoretical Astrophysics (graduate), with M. Bartelmann
- **Teaching assistant**, Universidad Autónoma de Madrid, 2013
  Gravitation and Cosmology; Physics of the Cosmos
- **Mini-course on inhomogeneous cosmologies**, University of Barcelona, 2012

Funding

- **ERC Consolidator Grant**, GLOW: Gravitational Lensing of Waves · 2,185,007 €
  Principal Investigator · European Research Council · 2026–2031
  Project site: glow-astro.org
- **Ramón y Cajal Fellowship**, ranked 2nd in Physics · IFT-UAM/CSIC · 2026–
  Atracción de Talento award · Agencia Estatal de Investigación · 100,000 €
  RyC-MaX excellence funding · CSIC · 225,000 €
- **Max Planck Society Annual Project**, With Einstein on Crooked Paths · 286,940 €
  Principal Investigator · funded by philanthropic donations · 2024–
- **Cosmology group core funding**, Max Planck Institute for Gravitational Physics
  Group Leader · Astrophysical and Cosmological Relativity division · 2020–
  Two postdocs and one PhD student

- **Workshop funding**, Lensing and Wave Optics in Strong Gravity · 12,000 €
  Erwin Schrödinger Institute, Vienna · 2024
- **Marie Skłodowska-Curie Global Fellowship**, NLO-CO · 264,668 €
  European Commission · UC Berkeley and IPhT Saclay · 2015–2020
- **FPI doctoral fellowship** · 54,816 € and 14,550 € for research stays
  Spanish Ministry of Science · stays in Heidelberg and Oslo · 2008–2012
- **Host** for two awarded Humboldt Fellowships (2022, 2025) and one Margarita Salas Fellowship (2022)

Other awards and fellowships

- R3 certification, Spanish Research Agency.
- Berkeley Connect Fellow, UC Berkeley (2018).
- Nordita Fellowship (2015).
- Yggdrasil Fellowship (2011): research stay at ITA Oslo.
- UAM Master's scholarship (2007).

Evaluation and peer review

Evaluation panels

- NASA Hubble Fellowship Program, Physics and Cosmology panel.
- FCT (Portuguese Foundation for Science and Technology), Individual Support Call.
- Max Planck Institute for Gravitational Physics, postdoctoral hiring committee, since 2020.

External evaluator for funding agencies

European Research Council (Starting Grants), STFC/UKRI, Swiss National Science Foundation, Alexander von Humboldt Foundation, InterTalentum (Marie Skłodowska-Curie COFUND), Netherlands Research Council (NWO), Research Grants Council of Hong Kong, and the Estonian Research Council.

Refereeing

Referee for journals including Physical Review Letters, Physical Review D, Nature Astronomy, JCAP, The Astrophysical Journal and Astrophysical Journal Letters.

Selected talks

- **Multi-Messenger Lensing Program**, Lorentz Center, Leiden, 2026 · plenary
  Don't kill the messenger! New phenomena, new challenges, new opportunities
- **Universidad Complutense de Madrid**, IPARCOS, 2026 · institute colloquium
  Gravitational lensing of waves
- **IAS and Princeton University**, gravitational-wave meeting, Princeton, 2026 · seminar
  GW231123 as a diffracted and magnified black-hole merger
- **Leibniz Institute for Astrophysics**, Potsdam, 2024 · institute colloquium
  Gravitational lensing of waves
- **New Frontiers in Strong Gravity**, Benasque, 2024 · plenary
  Wave-optical phenomena in gravitational-wave lensing
- **VIII Meeting on Fundamental Cosmology**, Granada, 2022 · plenary
  Gravitational-wave lensing and fundamental physics
- **EuCAPT Annual Symposium**, CERN, 2022 · plenary review
  Dark energy and tests of gravity with cosmology and gravitational-wave propagation

- **Progress on Old and New Themes in Cosmology**, Avignon (online), 2020 · plenary
  Tests of gravity with gravitational waves
- **COSMO**, Aachen University, 2019 · plenary
  Tests of dark energy and modified gravity
- **Rencontres du Vietnam**, Very High Energy Phenomena in the Universe, 2018 · plenary
  The dark Universe in the era of gravitational-wave astronomy

Languages

Spanish (native), English (C2), German (C1).

Selected publications

The complete record is on INSPIRE-HEP.

- Effective description of lensed gravitational waves diffracted by stellar fields
  M. Zumalacárregui, X. Shan · Phys. Rev. D 114, 063038 (2026) · Editors' Suggestion
- Across the Universe: GW231123 as a magnified and diffracted black hole merger
  S. Goyal, H. Villarrubia-Rojo, M. Zumalacárregui · Astrophys. J. Lett. 1008, L12 (2026)
- Black holes as telescopes: discovering supermassive binaries through quasi-periodic lensed starlight
  H. Wang, M. Zumalacárregui, B. Kocsis · Phys. Rev. Lett. 136, 061403 (2026) · Editors' Suggestion
- Gravitational wave lensing as a probe of halo properties and dark matter
  G. Tambalo, M. Zumalacárregui, L. Dai, M. Cheung · Phys. Rev. D 108, 103529 (2023)
- Gravitational wave lensing beyond general relativity: birefringence, echoes and shadows
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. D 102, 124048 (2020) · Editors' Suggestion
- Gravity in the era of equality: towards solutions to the Hubble problem without fine-tuned initial conditions
  M. Zumalacárregui · Phys. Rev. D 102, 023523 (2020)
- Limits on stellar-mass compact objects as dark matter from gravitational lensing of type Ia supernovae
  M. Zumalacárregui, U. Seljak · Phys. Rev. Lett. 121, 141101 (2018) · Editors' Suggestion
- Dark energy after GW170817: dead ends and the road ahead
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. Lett. 119, 251304 (2017) · Editors' Suggestion
- hi_class: Horndeski in the Cosmic Linear Anisotropy Solving System
  M. Zumalacárregui, E. Bellini, I. Sawicki, J. Lesgourgues, P. G. Ferreira · JCAP 08, 019 (2017)
- Transforming gravity: from derivative couplings to matter to second-order scalar–tensor theories beyond the Horndeski Lagrangian
  M. Zumalacárregui, J. García-Bellido · Phys. Rev. D 89, 064046 (2014)